

Clinical Risk Assessment Report: Patient 35

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Positive for Metastasis
- Prediction Confidence: High Confidence
- Absolute Predicted Risk: 41.4% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 1.87x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as High-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET), which elevated the patient's absolute risk by +6.0%. Biologically, this feature points to continuous, elongated bands of high-density or highly active tumor tissue.

2. Key Pathological & Imaging Drivers (Elevating Risk)

Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET) **Value: 0.7 [Top 13%] (+6.0%)**
Points to continuous, elongated bands of high-density or highly active tumor tissue.

Patient Age (Age) **Value: 46.0 [Bottom 5%] (+5.7%)**
Patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.

Squamous Cell Carcinoma Antigen (SCC_Ag) **Value: 0.41 [Bottom 10%] (+2.6%)**
Serum levels reflect tumor proliferation, burden, and squamous cell differentiation.

Small Zone Emphasis (PET) (GLZLM_SZE.PET) **Value: 0.16 [Avg (55th %ile)] (+2.5%)**
Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

75th Percentile Tumor Density (CONVENTIONAL_HUQ3) **Value: 0.65 [Avg (66th %ile)] (+2.3%)**
Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.

Large Zone Emphasis (CT) (GLZLM_LGZE.CT) **Value: -0.26 [Avg (35th %ile)] (+1.9%)**
Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

Short Run Low Gray-Level Emphasis (CT) (GLRLM_SRLGE.CT) **Value: -0.19 [Avg (26th %ile)] (+1.1%)**
Indicates fragmented areas of low density, often associated with diffuse micro-necrosis or loose tissue structure.

Tumor Sphericity (PET) (SHAPE_Sphericity(onlyFor3DROI)).PET) **Value: 1.39 [Top 3%] (+0.6%)**
Quantifies how closely the tumor resembles a perfect sphere; lower values indicate irregular, spiculated, or highly invasive margins.

3. Protective / Mitigating Factors (Reducing Risk)

No major risk-reducing features observed in the top drivers.

Machine Learning Feature Attribution (SHAP Decision Path)

